

Day 8 Report — Feature Engineering & Feature Selection

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Day: 8

Topic: Feature Engineering and Feature Selection

Dataset: Titanic Dataset

Tools & Libraries: Python, Pandas, NumPy, Matplotlib, Scikit-learn

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Objective

The objective of Day 8 was to:

- Understand the importance of Feature Engineering in Machine Learning.
 - Create meaningful features from existing dataset columns.
 - Apply Feature Selection techniques to identify the most informative features.
 - Compare model performance before and after feature engineering.
 - Analyze how engineered features impact predictive accuracy.
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Concepts Covered

Feature Engineering

Feature Engineering is the process of creating new features from existing data to improve machine learning model performance.

Features Created

1. **Title**
 - Extracted passenger titles (Mr, Mrs, Miss, Master, etc.) from the Name column.
2. **FamilySize**

Calculated using:

$\text{FamilySize} = \text{SibSp} + \text{Parch} + 1$

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3. **IsAlone**

- Indicates whether a passenger was traveling alone.

4. **FarePerPerson**

Calculated as:

$\text{FarePerPerson} = \text{Fare} / \text{FamilySize}$

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5. **AgeBand**

- Converted continuous age values into categorical age groups.
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Feature Selection

Feature Selection helps identify the most useful features while removing less informative ones.

Techniques Used

1. SelectKBest (ANOVA F-Test)

- Ranked features according to their statistical relationship with the target variable.
- Selected the top-performing features.

2. RFECV (Recursive Feature Elimination with Cross Validation)

- Iteratively removed the least important features.
 - Used cross-validation to determine the optimal number of features.
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Dataset Preparation

The Titanic dataset was preprocessed using the following steps:

- Missing Age values filled with median.
- Missing Embarked values filled with mode.
- Categorical variables encoded using Label Encoding.
- Features and target variable separated.

- Data split into training and testing sets.
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Baseline Model

A Random Forest Classifier was trained using only the original dataset features:

- Pclass
- Sex
- Age
- Fare
- Embarked

Baseline Accuracy

0.7877

Baseline Accuracy = 78.77%

Feature Engineered Model

The model was retrained after adding engineered features:

- Title
- FamilySize
- IsAlone
- FarePerPerson
- AgeBand

Engineered Model Accuracy

0.8268

Engineered Accuracy = 82.68%

Accuracy Comparison

Model	Accuracy
Baseline Model	78.77%
Engineered Model	82.68%

Improvement

82.68% - 78.77% = 3.91%

The engineered features improved the model accuracy by **3.91 percentage points**.

SelectKBest Results

Feature	Score
Sex	372.41
Pclass	115.03
Fare	63.03
FarePerPerson	45.91
IsAlone	38.35
Embarked	25.72
Title	4.26
Age	3.76
AgeBand	1.71
FamilySize	0.25

Top Features Identified

- Sex

- Pclass
 - Fare
 - FarePerPerson
 - IsAlone
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RFECV Results

Optimal Number of Features

5

Selected Features

- Sex
- Age
- Fare
- Title
- FarePerPerson

These features provided the best performance according to recursive feature elimination with cross-validation.

Visualizations Generated

The following graphs were created:

1. SelectKBest Feature Scores
2. RFECV Feature Selection Performance
3. Baseline vs Engineered Model Accuracy Comparison

These visualizations helped analyze feature importance and model improvement.

Key Observations

- Gender (Sex) was the most influential feature for predicting survival.
 - Passenger Class (Pclass) also had a strong relationship with survival.
 - The engineered feature FarePerPerson became one of the most informative features.
 - RFECV retained engineered features such as Title and FarePerPerson, proving their usefulness.
 - Feature Engineering significantly improved model performance without collecting additional data.
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Conclusion

During Day 8, meaningful features were engineered from the Titanic dataset and multiple feature selection techniques were applied. A Random Forest model trained on engineered features achieved an accuracy of 82.68%, compared to 78.77% for the baseline model. The experiment demonstrated that Feature Engineering and Feature Selection can substantially improve machine learning model performance by providing more informative inputs to the learning algorithm.

This task strengthened practical skills in data preprocessing, feature creation, statistical feature evaluation, recursive feature elimination, and model performance analysis.